

THIAGO SAUD

SENIOR AI & MLOPS ENGINEER | PRINCIPAL FRONT-END ENGINEER

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PROFESSIONAL SUMMARY

Principal Front-End Engineer with 7+ years architecting mission-critical web platforms in regulated financial ecosystems, uniting Software Architecture with end-to-end Engineering Governance. Reports directly to C-Level executives, conducting RFCs, ADRs and technical roadmaps in front of squads of up to 5 engineers. In active specialization in AI & MLOps Engineering - Python, Deep Learning, RAG, Agentic AI, LangChain/LangGraph and MLflow - consolidating a trajectory that unites product architecture, data engineering and applied artificial intelligence.

- **Designed** 10+ systems in production (trading, structured, advisors, hospitality, spatial BI), which support R\$1.6T+ in AUM/Custody and 4.8M+ active customers.
- **Boosted** growth of 37% YoY in revenue and ROAE of 28.1% through scalable architecture and technical governance.
- **Reduced** delivery time by 65% with AI-assisted workflows and shared libraries.
- **Raised** the quality/safety of perceived gain by ~80-90%, and code consistency (design systems, governance, tooling)

Engineering Philosophy: *Technical decisions treated as business decisions - architecture that scales products and people.*

EDUCATION

Bachelor of Computer Science | **Veiga de Almeida University (UVA) · Brazil, Rio de Janeiro** | JAN/2014 — FEV/2019

- **TCC - "Facial Recognition for Public Safety" (DEC/2019):** Designed and executed the complete design of the detection prototype (Viola-Jones) and facial recognition (Eigenfaces/PCA, Fisherfaces/LDA) using the Python language. Structured a real-time operational prototype that reached an accuracy of 80-100% and an average identification time of approximately 0.8 seconds (Python, OpenCV).

Executive Training in Administration and Business Management | **Barcelona University (UB) · Espanha, Catalunha** | JAN/2018 — NOV/2019

Continuing training in leader profile - business cases, decision-making and executive communication for technical leadership. Leader Profile program - business cases, decision-making and executive communication.

LICENSES AND CERTIFICATIONS

- **PUCRS — Professional and Emotional Skills (Leandro Karnal)** — Completed in AGO/2020
- **Google — Professional Specialization in UX & UI Design** — Completed in DEZ/2024
- **University of Michigan — Professional Specialization in Python** — Completed in AGO/2026
- **Anthropic — AI Fluency: Framework & Foundations** — Completed in AGO/2026
- **DeepLearning.AI — Professional Specialization in Deep Learning** — Completed in AGO/2026
- **IBM — Professional Certificate of AI Engineering** — In progress: goal OUT/2026
- **IBM — Professional Certificate RAG and Agentic AI** — In progress: goal NOV/2026
- **Duke University — Professional Specialization in MLOps | Machine Learning Operations** — In progress: goal DEZ/2026

LANGUAGES

Portuguese: Native | English: Technician | Spanish: Technician

CORE SKILLS

- **Technical Leadership and Architecture:** Software Architecture, RFCs, ADRs, Domain-Driven Design, Clean Architecture, SOLID, Micro-frontends (Module Federation), Design Systems, System Design, Monorepo.
- **Leadership and Team:** Hiring, Mentoring, Code Reviews, Performance Optimization, Cross-Functional Collaboration, Agile (Scrum · Kanban · RUP).
- **Product and Business:** Financial Services (Trading · Treasury · Brokerage), R\$1,6T+ AUM, 4,8M+ Active Users, 12K+ Advisors, Stakeholder Communication (C-Level).
- **AI Engineering:** Machine Learning, Deep Learning, Generative AI, NLP, Transformers, LLM/VLM Integration, RAG, LangChain, LangGraph, Agentic AI (CrewAI · MCP), Multimodal AI, Computer Vision (OpenCV), Vector Databases (FAISS · Chroma), Fine-tuning (LoRA · QLoRA), PyTorch, TensorFlow, Hugging Face, Scikit-learn, Prompt Engineering, Context Engineering, Harness Engineering, Responsible AI.
- **MLOps Engineering:** MLflow, AWS SageMaker, Azure ML, CI/CD for models, ONNX, Hugging Face Hub, GitHub Copilot.
- **Core Front-End:** Angular (Signals · NgRx · RxJS · Zoneless), React (Hooks · Redux · Redux-Saga · Context API), Vue.js, Next.js, HTML5, CSS3, SASS/LESS.
- **Core Programming Languages:** TypeScript, JavaScript (ES6+), Python, Java, C/C++.
- **Data, APIs & Rendering:** SQL, PostgreSQL, REST APIs, Data Visualization, GraphQL, Apollo Client, React Query, WebSockets, CSR/SSR/SSG/ISR, Virtualization, Lazy Loading.
- **Performance & Observability:** Core Web Vitals (LCP · INP · CLS), RUM/APM (Datadog · Sentry), Virtual Scroll, Pagination, GPU Rendering.
- **Security & Compliance:** OWASP Top 10, XSS, CSRF, JWT, 2FA, SSO, OIDC, OAuth 2.0, TLS, CSP.
- **Delivery & DevOps:** CI/CD (Azure DevOps · GitHub Actions), Docker, AWS (S3 · CloudFront), SonarQube, Playwright, Jest, Husky.
- **UX/UI & Design Systems:** User-Centered Design, Wireframing, Prototyping, Usability Testing, BEM, Responsive Design, WCAG/A11Y, Storybook, Figma.

WORK EXPERIENCES

PRINCIPAL FRONT-END ENGINEER | BTG Pactual Bank S.A.

FEB/2023 — JUN/2026 | São Paulo, Brazil (Remote)

Served as a Specialist and Technical Leader in the largest investment bank in Latin America, vertically Digital Equities, with direct reporting to the Managing Partner and the Executive Director. He led engineering governance through RFCs and ADRs, in addition to managing the mentoring of 2 to 5 engineers. Conducted the pioneering adoption of AI Engineering (LLMs/VLMs via MCP, Cursor and Copilot), resulting in a 65% reduction in delivery time and an 80% increase in code quality. The platforms under its architecture serve 4.8M+ active customers and manage R\$1.6T+ in AUM/Custody, supporting high-frequency and low latency operations in B3, which boosted the growth of 37% YoY in revenue, ROAE of 28.1% (3Q2025) and the historical cost-to-income record of 34.1%.

Strategic Products:

- **Home Broker Web (White-Label):** Designed, coordinated and executed the design of the digital trading platform aimed at 4.8M+ active customers, optimized for high frequency of real-time operations in B3 and under strict regulatory compliance criteria.
- **COE (Certificate of Structured Operations):** Designed and executed the design of the platform of complex fixed income and derivative financial products, using a micro-frontend structure integrated with advanced authentication flows and regulatory compliance.
- **Internal Corporate Library:** Designed and coordinated the design and governance of the library of reusable and standardized Angular components for the entire Digital Equities vertical, directly impacting the bank's operational cost efficiency.

Architecture & Engineering:

- **Software Architecture:** Designed, coordinated and executed the orchestration layer to isolate business rules from the UI. (*Angular Signals, Zoneless, DDD, Clean Architecture, SOLID*)
- **Distributed Micro-frontends:** Structured independent deploys and loading on demand by domain. (*Webpack Module Federation*)
- **Performance & Reactive State:** Designed, coordinated and executed the reactive layer to sustain peaks of high volatility in the financial market without UI degradation, reducing CPU usage and infrastructure costs. (*NgRx, Virtual Scroll, Pagination, Defer, Debounced, QueueMicrotask, GPU rendering*)
- **Security & Compliance:** Established the hybrid session architecture and proactive XSS mitigation architecture in a highly sensitive regulated environment. (*OWASP Top 10, DomSanitizer, Cookies HttpOnly/SameSite*)
- **Observability & Cloud:** Implemented the continuous monitoring ecosystem to identify regressions in production with low latency global distribution. (*Datadog, Sentry, Lighthouse, AWS S3/CloudFront*)
- **Design System (ORCHESTRA):** Coordinated corporate standardization that accelerated the delivery time of product fronts and ensured strict compliance with WCAG/A11Y standards. (*Storybook, Nx/Turborepo, Nexus*)
- **CI/CD & Quality:** Configured automated guardrails to ensure code stability even under deploy stress scenarios. (*Azure DevOps, GitHub Actions, Husky, SonarQube, Karma, Playwright*)

SENIOR FRONT-END ENGINEER | XP Investments S.A.

JUL/2021 — NOV/2022 | São Paulo, Brazil (Remote)

Served as a Specialist and Technical Leader at the largest investment broker in Latin America, Vertically Global Markets, with direct reporting to the Engineering Manager and the Director of Operations/CFO/Treasury, developing platforms aimed at the Operations Desk and for more than 12,000 Investment Advisors, serving about 4.5M active customers. He collaborated strategically with the Sales, Advisors, Design and Backend teams, contributing to the governance of platforms that manage R\$1.2T in total assets and drive R\$19.8 billion in gross revenue.

Strategic Products:

- **Black Sales (Operations Table):** Designed and executed the design of the system platform for registration and distribution of structured and derivative products, supporting the management of R\$1.2T in total assets.
- **Black AAI (Advisors):** Designed and executed the design of the profile analysis, recommendation and portfolio management platform used by ~12 thousand advisors to serve ~3.8M active customers.

Architecture & Engineering:

- **Software Architecture:** Designed high-performance dynamic interfaces, decoupling business transactional rules from the UI layer. (*React, DDD, Docker, Clean Architecture, SOLID*)
- **Distributed Micro-frontends:** Structured architecture based on independent deploys and on-demand loading by corporate domain. (*Webpack Module Federation*)
- **Data & State:** Optimized the synchronization of complex custody and wallet data without overloading the backend infrastructure. (*React Query, GraphQL, Apollo Client, Redux-Saga*)
- **Rendering Performance:** Mitigated processing bottlenecks and loss of graphic frames during processing critical scenarios of sending high-frequency orders. (*React Hooks, Virtual Scroll, Pagination, GPU rendering*)
- **Security & Observability:** Implemented input sanitization routines and continuous monitoring of regressions with resilient distribution of static assets. (*OWASP Top 10, Datadog, Sentry, AWS S3/CloudFront*)
- **Design System (SOMA):** It integrated corporate standardization that drastically reduced the delivery time of product fronts and ensured strict compliance with WCAG/A11Y standards. (*Storybook, Nx/Turborepo, Nexus*)
- **CI/CD & Quality:** Configured automated guardrails on deployment mats to ensure code resilience and stability under market stress scenarios. (*Azure DevOps, GitHub Actions, Husky, SonarQube, Jest, Playwright*)

MID-LEVEL FRONT-END ENGINEER | Zoox Technology Ltda.

JUL/2020 — NOV/2020 | Rio de Janeiro, Brazil (Remote)

Served as a technical reference and Systems Analyst on a high-complexity white-label PMS platform, performing direct dialogue with senior client stakeholders. Integrated real-time billing with global OTAs (Booking.com, Expedia, Agoda), eliminating data competition failures and overbooking. Modeled and implemented the end-to-end detection and facial recognition flow for user validation.

- **Digital Guest Journey:** Developed the Check-in and Check-out flow based on asynchronous components, reducing operational waiting time and eliminating bottlenecks in access peaks. (*Vue.js, Vuetify, Docker, PostgreSQL*)
- **Facial Recognition (Amazon Rekognition):** Designed the flow of face detection (bounding boxes) and recognition (face matching with trust score) for biometric validation, ensuring strict compliance with the FNHR. (*AWS Rekognition, TLS, OWASP Top 10*)
- **In-Room Monetization:** Structured transactional module with centralized state management and instant synchronization, driving the increase in auxiliary revenue of establishments. (*Vuex*)

FRONT-END ENGINEER | TG4 Technology Ltda.

SEP/2019 — JUL/2020 | Rio de Janeiro, Brazil (Presencial)

Served as a Systems Analysis with direct reporting to the Co-Founder and CTIO. Planned, designed and executed the white-label chatbot platform aimed at the largest telecommunications operators in the country (Tim, Claro, Oi, Vivo, Nextel). The product integrated predictive recommendation algorithms that generated a 100% increase in the acceptance of upgrade offers, in addition to drastic reduction of CAC/CPC and maximization of LTV.

- **Software Architecture:** Designed and executed the orchestration layer to isolate complex business rules from the UI layer. *(Angular, Zone.js, Docker, DDD, Clean Architecture, SOLID)*
- **Security & Governance:** Implemented input sanitization routines and continuous monitoring of regressions with resilient distribution of static assets. *(OWASP Top 10, Datadog, Sentry, AWS S3/CloudFront)*
- **CI/CD & Quality:** Configured automated guardrails for integration and deploy stress scenarios. *(Azure DevOps, Husky, SonarQube, Karma, Playwright)*

INTERN IN RESEARCH AND TECHNOLOGICAL DEVELOPMENT (R&D) | Neoris of Brasil Ltda.

APR/2017 — APR/2019 | Rio de Janeiro, Brazil (Híbrido)

Served as a the sole intern in the Engineering Division of Research and Technological Development (R&D), reporting to three Senior Business Managers, focusing on spatial computing, geospatial BI and cognitive prototypes of AI (facial recognition, AR/VR). Planned, designed and executed two strategic products for the Aliansce client and two internal PoCs, applying POO and validation/escape of payloads based on OWASP guidelines from the conceptual phase.

Strategic Products:

- **Aliansce Web (Multi-story Mapping and Metrics Analysis):** Designed and executed the design of the web platform without frameworks focused on spatial BI of malls, crossing billing indicators per store on an interactive map. *(Javascript Vanilla OOP, Google Maps API, Google Drawing API, SVG dinâmico)*
- **Aliansce Mobile (AR/VR - Real-time Spatial Metrics Visualization):** Designed and executed the design of the native android space computing app focused on 3D tracking and optical target recognition for overlay of operational indicators in augmented reality. *(Java, Android Studio, Vuforia SDK, Geolocation)*
- **Neoris Mobile (AR/VR - Real-time Spatial Metrics Visualization):** Designed and executed the design of the prototype android app native to space computing focused on executive presentations, performing the volumetric rendering of 3D dashboards powered by BI in real time. *(Java, Android Studio, Vuforia SDK, Geolocation)*
- **Neoris Web (AI Vision - Media Ingestion and Cognitive Classification):** Designed and executed the design of the web prototype with an emphasis on asynchronous media upload, automating object detection, people tracking, demographic classification and content moderation, consolidating the fundamentals in Computer Vision and AI Engineering. *(Javascript Vanilla, Microsoft Azure Cognitive Services)*